The first part is the basic theory of Banach space. This is an extra contents for the functional analysis.

Chapter 1 Banach Spaces

1.1 Definitions, and Basic Properties

Definition 1.1.0.1 A normed space $(X, \|-\|)$ is called a **Banach space** if it is compelete as a metric space.

1.1.1 Banach Space from Normed Space: Compeletion

♠ **Proposition 1.1.1.1** ...

∘ **Proof** ... □

Chapter 2 Hilbert Spaces

Recall:
◇ **Lemma 2.0.0.1** (Jordan-von Neumann)TFAE:

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∘ **Proof** ... □

2.1 Pre-Hilbert Space

2.2 Hilbert Space

Definition 2.2.0.1 A **Hilbert space** is a complex linear space which is complete in the metric induced by the norm.
i.e. it is a complex Banach space.

Equivalently, it is equipped with a compelete complex inner product.

2.2.1 Duality and Representation

Riesz's Representation Theorem

Self-Duality

Reflxivity

Riesz-Frechet

Lax-Milgram Theorem

Chapter 3 Operators on Hilbert Spaces

The second part is the Banach space theory.

The third part is the Interpolation Theory.

Chapter 4 Classical Interpolation Theory

Chapter 5 Absolute Interpolation Theory